MTH166

Lecture-23

Solution of Partial Differential Equations

And

Solution of 1D-Heat Equation

Topic:

Solution of Partial Differential Equations

Learning Outcomes:

To solve first order PDE using separation of variables solution.

To solve one dimensional Heat Equation.

Find the separation of variables solution of following PDE:

Problem 1. $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y}$

Solution. The given equation is: $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y}$ (1)

Let solution be: $u(x, y) = XY$ (2) where $X = f(x), Y = g(y)$

$\Rightarrow \frac{\partial u}{\partial x} = X'Y$ and $\frac{\partial u}{\partial y} = XY'$

Equation (1) becomes: $X'Y = XY'$

$\Rightarrow \frac{X'}{X} = \frac{Y'}{Y} = k$ (Say)

Taking these pairs one by one

$\Rightarrow \frac{X'}{X} = k$

$$\Rightarrow \int \frac{X'}{X}\,dx = k\int dx$$

$$\Rightarrow \log X = kx + c_1$$

$$\Rightarrow X = e^{(kx+c_1)}$$

also $\frac{Y'}{Y} = k$

$$\Rightarrow \int \frac{Y'}{Y}\,dy = k\int dy$$

$$\Rightarrow \log Y = ky + c_2$$

$$\Rightarrow Y = e^{(ky+c_2)}$$

Required solution is: $u(x,y) = XY = e^{(kx+c_1)}e^{(ky+c_2)} = e^{(c_1+c_2)}e^{k(x+y)}$

$\Rightarrow u(x,y) = Ae^{k(x+y)}$ **Answer.** $(A = e^{(c_1+c_2)})$

Find the separation of variables solution of following PDE:

Problem 2. $4\frac{\partial u}{\partial x} + 3\frac{\partial u}{\partial y} = 0$

Solution. The given equation is: $4\frac{\partial u}{\partial x} + 3\frac{\partial u}{\partial y} = 0$ (1)

Let solution be: $u(x, y) = XY$ (2) where $X = f(x), Y = g(y)$

$\Rightarrow \frac{\partial u}{\partial x} = X'Y$ and $\frac{\partial u}{\partial y} = XY'$

Equation (1) becomes: $4X'Y + 3XY' = 0$ $\Rightarrow 4X'Y = -3XY'$

$\Rightarrow 4\frac{X'}{X} = -3\frac{Y'}{Y} = k$ (Say)

Taking these pairs one by one

$\Rightarrow 4\frac{X'}{X} = k$

$$\Rightarrow \int \frac{X'}{X} dx = \frac{k}{4} \int dx$$

$$\Rightarrow \log X = \frac{k}{4} x + c_1$$

$$\Rightarrow X = e^{\left(\frac{k}{4}x + c_1\right)}$$

also $-3\frac{Y'}{Y} = k$

$$\Rightarrow \int \frac{Y'}{Y} dy = -\frac{k}{3} \int dy$$

$$\Rightarrow \log Y = -\frac{k}{3} y + c_2$$

$$\Rightarrow Y = e^{\left(-\frac{k}{3}y + c_2\right)}$$

Required solution is: $u(x, y) = XY = e^{\left(\frac{k}{4}x + c_1\right)} e^{\left(-\frac{k}{3}y + c_2\right)} = e^{(c_1 + c_2)} e^{k\left(\frac{1}{4}x - \frac{1}{3}y\right)}$

$\Rightarrow u(x, y) = Ae^{\frac{k}{12}(3x - 4y)}$ **Answer.** $\left(A = e^{(c_1 + c_2)}\right)$

Find the separation of variables solution of following PDE:

Problem 3. $\mathbf{y}\frac{\partial \boldsymbol{u}}{\partial \boldsymbol{x}} + \boldsymbol{x}\frac{\partial \boldsymbol{u}}{\partial \boldsymbol{y}} = \mathbf{0}$

Solution. The given equation is: $y\frac{\partial u}{\partial x} + x\frac{\partial u}{\partial y} = 0$ (1)

Let solution be: $u(x,y) = XY$ (2) where $X = f(x), Y = g(y)$

$\Rightarrow \frac{\partial u}{\partial x} = X'Y$ and $\frac{\partial u}{\partial y} = XY'$

Equation (1) becomes: $yX'Y + xXY' = 0 \quad \Rightarrow yX'Y = -xXY'$

$\Rightarrow \frac{1}{x}\frac{X'}{X} = -\frac{1}{y}\frac{Y'}{Y} = k$ (Say)

Taking these pairs one by one

$\Rightarrow \frac{1}{x}\frac{X'}{X} = k$

$$\Rightarrow \int \frac{X'}{X} dx = k \int x dx$$

$$\Rightarrow \log X = k\frac{x^2}{2} + c_1$$

$$\Rightarrow X = e^{\left(k\frac{x^2}{2}+c_1\right)}$$

also $-\frac{1}{y}\frac{Y'}{Y} = k$

$$\Rightarrow \int \frac{Y'}{Y} dy = -k \int y dy$$

$$\Rightarrow \log Y = -k\frac{y^2}{2} + c_2$$

$$\Rightarrow Y = e^{\left(-k\frac{y^2}{2}+c_2\right)}$$

Required solution is: $u(x,y) = XY = e^{\left(k\frac{x^2}{2}+c_1\right)}e^{\left(-k\frac{y^2}{2}+c_2\right)} = e^{(c_1+c_2)}e^{k(\frac{x^2}{2}-\frac{y^2}{2})}$

$\Rightarrow u(x,y) = Ae^{\frac{k}{2}(x^2-y^2)}$ **Answer.** $(A = e^{(c_1+c_2)})$

Polling Quiz

If $u = f(x,t)$, then the solution will be of the form:

(A) $u(x,y) = X(x)Y(y)$

(B) $u(x,t) = X(x)T(t)$

(C) $u(x,y) = X(t)Y(x)$

<u>Problem.</u> Solve $\frac{\partial u}{\partial t} = C^2 \frac{\partial^2 u}{\partial x^2}$ [1D-Heat Equation]; c^2 is Diffusivity constant.

<u>Solution.</u> The given one dimensional Heat equation is:

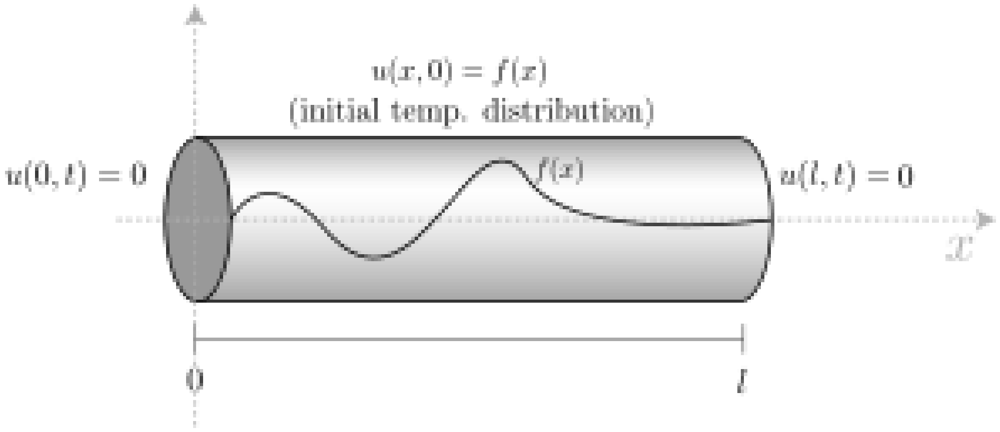
$$\frac{\partial u}{\partial t} = C^2 \frac{\partial^2 u}{\partial x^2} \qquad (1)\ 0 \le x \le l\ (length), t > 0\ (time)$$

Let solution be: $u(x,t) = XT$ (2) where $X = f(x), T = g(t)$

$$\Rightarrow \frac{\partial u}{\partial t} = XT' \quad \text{and} \quad \frac{\partial^2 u}{\partial x^2} = X''T$$

Equation (1) becomes: $XT' = C^2 X''T$

$$\Rightarrow \frac{X''}{X} = \frac{1}{C^2}\frac{T'}{T} = k \quad \text{(Say)}$$



As k can take three values: zero, positive or negative, so we have following three cases.

Case 1. When $k = 0$

$$\frac{X''}{X} = k \qquad \Rightarrow \frac{X''}{X} = 0 \qquad \Rightarrow X'' = 0 \qquad \Rightarrow X = ax + b$$

Also $\frac{1}{C^2}\frac{T'}{T} = k \qquad \Rightarrow \frac{1}{C^2}\frac{T'}{T} = 0 \qquad \Rightarrow T' = 0 \qquad \Rightarrow T = c$

Required solution of equation (1) is:

$$u(x,t) = XT = (ax + b)c = Ax + B$$

Case 2. When $k = p^2$ (Positive)

$$\frac{X''}{X} = k \qquad \Rightarrow \frac{X''}{X} = p^2 \qquad \Rightarrow X'' - p^2X = 0$$

S.F. $(D^2 - p^2)X = 0$

A.E. $(D^2 - p^2) = 0 \qquad \Rightarrow D = \pm p$

$\therefore X = ae^{px} + be^{-px}$

Also $\frac{1}{C^2}\frac{T'}{T} = k \quad \Rightarrow \frac{1}{C^2}\frac{T'}{T} = p^2 \quad \Rightarrow T' - C^2p^2T = 0$

S.F. $(D - C^2p^2)T = 0$

A.E. $(D - C^2p^2) = 0 \quad \Rightarrow D = C^2p^2$

$\therefore T = ce^{C^2p^2t}$

Required solution of equation (1) is:

$$u(x,t) = XT = (ae^{px} + be^{-px})\left(ce^{C^2p^2t}\right) = (Ae^{px} + Be^{-px})e^{C^2p^2t}$$

Case 3. When $k = -p^2$ (Negative)

$\frac{X''}{X} = k \quad \Rightarrow \frac{X''}{X} = -p^2 \quad \Rightarrow X'' + p^2X = 0$

S.F. $(D^2 + p^2)X = 0$

A.E. $(D^2 + p^2) = 0 \qquad \Rightarrow D = \pm ip$

$\therefore X = e^{0x}(a\cos px + b\sin px)$

Also $\quad \frac{1}{C^2}\frac{T'}{T} = k \qquad \Rightarrow \frac{1}{C^2}\frac{T'}{T} = -p^2 \qquad \Rightarrow T' + C^2p^2T = 0$

S.F. $(D + C^2p^2)T = 0$

A.E. $(D + C^2p^2) = 0 \qquad \Rightarrow D = -C^2p^2t$

$\therefore T = ce^{-C^2p^2t}$

Required solution of equation (1) is:

$$u(x,t) = XT = (a\cos px + b\sin px)\left(ce^{-C^2p^2t}\right) = (A\cos px + B\sin px)e^{-C^2p^2t}$$

This is the most suitable and practically feasible solution of heat equation.

Note: Heat Equation

Equation: $\frac{\partial u}{\partial t} = C^2 \frac{\partial^2 u}{\partial x^2}$

Nature: Parabolic

Solution: **1.** $u(x,t) = Ax + B$

2. $u(x,t) = (Ae^{px} + Be^{-px})\, e^{C^2p^2t}$

or

$u(x,t) = (A\cosh px + B\sinh px)e^{C^2p^2t}$

3. $u(x,t) = (A\cos px + B\sin px)e^{-C^2p^2t}$ **(Most suitable one)**